NATIONAL UNIVERSITY OF SINGAPORE

CS5224 – Cloud Computing

(Semester 1: AY 2025/2026)

Time Allowed: 1.5 Hours

Sample Answers

INSTRUCTIONS TO STUDENTS

1. This assessment paper contains **TWENTY FIVE (25)** questions.
2. Total marks in this question paper is **30**.
3. Please **ALL** questions.
4. This is a **NON-SECURE BLOCK INTERNET** assessment.
5. This is an **OPEN BOOK** assessment. You are allowed to get reading materials as well as use study materials on your laptop.

Examiner's Use Only

Question No.	Maximum Marks
MCQs 1 – 20	20
21	2
22	2
23	2
24	2
25	2
Total	30

Question 1

What is a key benefit of using cloud-based databases compared to traditional on-premise non-cloud databases?

(a) Increased security
(b) Lower upfront costs
(c) Increased accessibility and scalability
(d) Cheaper storage
(e) Lower latency

Option (c). (a) and (d) are not guaranteed.

Question 2

You are the founder of a start-up with very limited personnel. Your company handles the input and processing of infectious disease clinical trial data on the cloud. The data processing needs to be real-time. The data contain highly sensitive, personally identifiable information. However, you want to reduce as much manual work as you can due to your small team. Which service should you choose for this application?

(a) IaaS
(b) FaaS
(c) CaaS
(d) PaaS
(e) SaaS

Option (d). SaaS gives the least amount of manual work, but the data processing cannot be real-time, and data privacy is also a concern. Therefore, PaaS is the best solution for minimizing work while meeting privacy and real-time requirements.

Question 3

Which one of the statements below about the number of workers in a MapReduce framework is correct?

(a) Too many workers can reduce overall efficiency because it overloads the master node's task and increases data transfer overhead between workers
(b) The number of workers has no effect on performance, because MapReduce automatically balances tasks perfectly
(c) Using fewer workers than available CPUs is always better, because it reduces network usage and eliminates the risk of data skew
(d) MapReduce performance depends only on the size of the input data, not on the number of workers assigned
(e) Having too few workers is not efficient because total worker idle time increases

Option (b) is incorrect as more workers usually reduces latency up to a certain point. (c) is incorrect because some CPUs will be idle and network is not the only overhead. (d) is incorrect because number of workers does determine latency. (e) is incorrect because

total worker idle time does not increase a worker's idle time in most cases. Fewer workers mean each will get more tasks, which decreases their idle time.

Question 4

You have an idea for an app that finds local festivity celebration parties. During the few festivities throughout the year (e.g. New Year, National Day, etc.) many users will use your app to find the most suitable party for them to go to based on users' criteria. Which AWS service is the most suitable to the usage profile and also cost effective?

(a) EC2: Virtual server to run your app
(b) S3: Store static app contents (party flyers, images) and serve directly to users
(c) Elastic Beanstalk: Managed EC2 with auto-scaling
(d) CloudFront: Cache contents and reduce latency for app users
(e) Lambda: App logic as a serverless function

Option (e). Traffic will be bursty during festive periods but not otherwise. Lambda does not incur continuous cost when app is idle, so it is cost-effective. EC2 and EBS incur continuous cost even when app is idle. CloudFront and S3 are only for static content delivery but does not handle app logic or compute (e.g. which one is the best party for me?)

Question 5

According to the MapReduce execution flow described in the slides, which statement most accurately describes the data flow and the division of responsibility between the developer and the system?

(a) The developer codes the Map function, which writes intermediate data to a shared storage server, and the Reduce function, which reads from that shared storage to aggregate results
(b) The developer codes the Map and Reduce functions, but must also explicitly implement the "Shuffle and Sort" logic to ensure intermediate data is correctly grouped for the reducers
(c) The Master instance partitions the input data, assigns segments to Map instances, and then directly sends the intermediate output from each Map instance to the correct Reduce instance
(d) The developer's Map function writes intermediate results to the local disk of its worker node; the MapReduce system is then responsible for shuffling, merging, and sorting this data before feeding it to the developer's Reduce function
(e) Map instances run in parallel, and Reduce instances run in parallel. To optimize performance, a Reduce worker is always placed on the same node as the Map worker that generated its data, eliminating network shuffling

Answer follows directly from bookwork

Question 6

Which one of the following statements most accurately distinguishes the architectural difference between Virtualization (using a Type 1 Hypervisor) and Multitenancy?

(a) Virtualization and Multitenancy are identical concepts, both referring to a single software instance serving multiple tenants
(b) Virtualization involves a hypervisor running on a Host OS, while Multitenancy involves a hypervisor running directly on the physical infrastructure
(c) Virtualization is an enabling technology that abstracts physical hardware (e.g., CPU, memory), allowing a single server to run multiple, isolated guest OSs. Multitenancy is a software architecture where a single instance of an application serves multiple tenants, keeping their data and configuration isolated
(d) Virtualization is a security risk due to Guest OS attacks, whereas Multitenancy is inherently secure because it always guarantees tenant isolation and data security
(e) Virtualization is used for IaaS, while Multitenancy is an architecture used exclusively for SaaS

Answer follows directly from bookwork

Question 7

Assuming you are moving from an on-premise deployment to a cloud deployment, which is the most important area of concern?

(a) Scalability
(b) Security
(c) Load Balancing
(d) Storage
(e) Bandwidth

Answer: Security. Rest are dependent on SLAs and what you can get, but by going to the Cloud, your data is no longer under your control to some extent

Question 8

You are running a cloud environment and realise you need a load balancer. What is its primary purpose?

(a) To ensure a high bandwidth between your application and the client
(b) To automatically adjust the necessary amount of computing resources that are used by your applications
(c) To distribute incoming service requests across multiple servers
(d) To ensure that data in the cloud is synchronised across multiple servers
(e) To manage the various computing resources within a datacenter

Option (c)

Question 9

A datacenter operator is analysing their energy consumption using the "Energy Usage and Power Efficiency" pie chart. They want to improve their Power Usage Efficiency (PUE), which is calculated as:

$$\left(\frac{\text{total power used by datacenter}}{\text{power used by IT equipment}}\right)$$

Total power used by datacenter includes both equipment and overhead power. To achieve an ideal PUE of 1, their efforts should focus on reducing the power consumption of which components?

(a) CPUs, DRAM, and Disks
(b) Power Overhead and Cooling Overhead
(c) Networking and Miscellaneous
(d) CPUs
(e) The Electrical Yard and Main Server Hall

Option (b). Options (a), (c), (d) are IT equipment, so reducing their power will reduce both the numerator and denominator, making it impossible to reach PUE of 1. (e) is wrong because main server hall contains CPUs, RAM, Disks, etc. too

Question 10

A cloud-native consulting firm offers a new product to its clients. This product integrates a SaaS-based Cloud Resource Management from Provider A, a PaaS-based database from Provider B, and a Storage-as-a-Service solution from Provider C. The firm combines these into a single, new, unified service with one dashboard and one bill, and it manages the secure data movement between the three providers. According to the NIST Cloud Computing Reference Architecture, which specific role and service is this firm fulfilling?

(a) Cloud Auditor, performing a Security Audit
(b) Cloud Carrier, providing transport of cloud services
(c) Cloud Broker (Service Intermediation)
(d) Cloud Broker (Service Arbitrage)
(e) Cloud Broker (Service Aggregation)

Option (e)

Question 11

An organization has deployed a redundant service (Service A) on three virtual servers (Server A, Server B, Server C). To manage load, all consumer requests are sent *directly* to Server A. Server A contains application logic that allows it to assess its own load and forward a portion of the incoming requests to Server B and Server C as needed. This differs from a model where an external device intercepts all traffic. Which specific architecture does this scenario illustrate?

(a) Independent Load Balancer
(b) Sibling Resource Pool
(c) Dynamic Vertical Scaling
(d) Built-In Load Balancer
(e) Cloud Bursting Architecture

Option (a) is incorrect because independent load balancer handles all incoming requests before it reaches server.

Question 12

You are part of an e-commerce start-up and wish to understand the shopping patterns arising from last year's sales events. In order to do this, you have been tasked with doing the data analysis where you have a lot of data. Which cloud deployment will you choose as the best option?

(a) Public cloud only
(b) Private cloud only
(c) Hybrid cloud
(d) Either Public or Private cloud
(e) Multi-cloud

Option (d) as either would be fine depending on your needs.

Question 13

Who amongst the following are primary users of PaaS?

(a) Clients looking to use a software online
(b) Software developers
(c) System administrators
(d) Hardware engineers
(e) Network engineers

Option (b)

Question 14

A company wants to move its deployment to the cloud but is worried about the security threats in the cloud. Which one of the following is the major security threat that the company needs to worry about?

(a) Unauthorized physical access
(b) Data breaches due to multi-tenancy
(c) Outdated antivirus software
(d) Limited user authentication methods
(e) Data leaks due to bad programming practices

Option (b)

Question 15

Which one of the following statements is NOT true about Cloud Computing?

(a) Opex includes expenses like electricity and maintenance for on-premise IT resources
(b) Detecting spam in social media is a good example of a use case for MapReduce
(c) PaaS is the cloud service that is usually provided to run the development environment
(d) Elastic scaling can dynamically increase or decrease server resources based on a specific period
(e) Higher security threats are a disadvantage of using public PaaS for SaaS development

Option (d). There is no specific time period as this depends on the load

Question 16

Your company has a policy of encrypting all data that is stored. Your data is hosted in a virtual private cloud and attached to it are multiple storage devices which has the data that must be encrypted. You have the following options available. Based on this, which one of the following options will allow you to do this? (if multiple options are possible choose the one that allows you the most choice)

(i) Ask clients to encrypt their data before they give it to you
(ii) Use third party encryption tools to encrypt the data
(iii) Use native encryption tools available in the Operating System to encrypt the data
(iv) Encrypt your data within the application before storing

(a) (i) or (iii)
(b) Only (iv)
(c) (ii) or (iii) or (iv)
(d) (i) or (ii) or (iii)
(e) (iii) or (iv)

Option (c). All of these are fine. Asking clients is risky in case they forget ☺

Question 17

In a cloud environment, which type of scaling adjusts the number of resources based on the load and traffic?

(a) Load scaling
(b) Vertical scaling
(c) Horizontal scaling
(d) Network scaling
(e) Elastic scaling

Option (e).

Question 18

You realise that using containerised applications in a datacenter environment gives you certain advantages. Which one of the following could be one of them?

(a) Increases the physical size of the server infrastructure
(b) Eliminates the need for cloud-based applications
(c) Improves isolation between applications and operating system
(d) Reduces the number of physical servers needed
(e) Reduces the overall cost

Option (c).

Question 19

When expecting a huge variation in traffic, as a cloud engineer, we need to be able to ensure the application can scale dynamically based on demand. Which cloud computing service model would we use to implement scalability?

(a) Elastic load balancer
(b) SaaS
(c) Auto-scaling
(d) PaaS
(e) None of the above

Option (d) as it provides scaling. (a), (c) are not service models.

Question 20

Which of the following best distinguishes Dynamic Scalability Architecture from Elastic Resource Capacity Architecture?

(a) Dynamic Scalability relies on manual scaling operations, whereas Elastic Resource Capacity is fully automated and static
(b) Dynamic Scalability only involves vertical scaling, while Elastic Resource Capacity only involves horizontal scaling
(c) Dynamic Scalability focuses on scaling within a single resource pool, whereas Elastic Resource Capacity focuses solely on scaling between sibling pools
(d) Dynamic Scalability requires an Automated Scaling Listener, while Elastic Resource Capacity does not use any monitoring component
(e) Dynamic Scalability adjusts resources based on predefined conditions, while Elastic Resource Capacity uses intelligent automation to allocate or reclaim resources in real time

Option (e) – from the notes

Question 21

AWS provides instances that are optimized for different workloads. Some instances are memory-optimized, others are I/O optimized. You are assigned a project to be run on AWS. The project requires 60 Peta FLOPs and 800,000 I/O operations to complete. Assume that compute and I/O operation are sequential. The details of each instance are provided below. Which instance should you choose to optimize your project cost? Please show your working.

Tera = 10^{12}. Peta = 10^{15}. FLOP = Floating Point Operations. ms = 10^{-3} seconds.

Name	Computation Speed (TeraFLOP/s)	I/O Latency (ms/IOops)	Price ($/sec)
m7 (General purpose)	1	3	1
i8 (I/O optimized)	1	1	2
c5a (Compute optimized)	3	2	2
r7 (Memory optimized)	1	3	2
g4 (GPU)	30	10	5

Answer:

Name	TeraFLOP/s	I/O Latency (ms)	Price ($/sec)	N_ops (TeraFLOP)	N_I/O_ops	Total Time (s)	Price ($)
m7	1	3	1	60000	800000	62400	62400
i8	1	1	2	60000	800000	60800	121600
c5a	**3**	**2**	**2**	**60000**	**800000**	**21600**	**43200**
r7	1	3	2	60000	800000	62400	124800
g4	30	10	5	60000	800000	10000	50000

Question 22

As part of an annual review in your company, you realise you have been with your cloud provider for many years and start to worry about the potential risks of 'vendor lock-in'. List **two** methods in which you could help mitigate this risk. Briefly (in one or two sentences) explain your answer.

Answer: less use of proprietary technologies and hence potentially easier to migrate, spread deployment across multiple providers, using technologies such as Docker to make it easier to move workloads across different environments, etc.

Question 23

It is the upcoming holiday season and your application has gained a lot of traction – yay! However, the downside is that your cloud infrastructure is struggling with this load. You have looked into auto-scaling and load-balancing already. List **two** other methods that you can use to help with this issue.

Answer: Use caching, CDNs, optimise database for faster query processing, etc.

Question 24

You are responsible for the cloud infrastructure in company WeAreAwesome Inc. As awesome as you are, you have however noticed that your VMs that are running on the cloud are costing more than they should. Given your understanding of Cloud Computing, you wish to optimise the cost of running the VMs. Give **two** ways in which you can achieve this.

Answer: Try to eliminate over-provisioning, use reserved instances for predictable workloads, or spot instances for non-critical workloads , can do auto-scaling to scale-down as necessary

Question 25

While setting up a new cloud deployment, you are tasked with ensuring that data availability must be high and that you must also comply with data protection guidelines which mandate that you must store Singapore customer data only within Singapore. How can you ensure compliance in such a scenario while keeping costs down? Briefly explain your answer.

Answer: Use a sovereign cloud or select specific regions for data storage and ensure that data replication occurs only within those regions.

– END OF PAPER –